

EDWARD BAKER

MACHINE LEARNING ENGINEER | SOFTWARE ENGINEER | DATA SCIENTIST

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Production-focused engineer building machine learning models, cloud pipelines, APIs, and reusable Python systems.

TECHNICAL SKILLS

Languages: Python, SQL, Java, C/C++, JavaScript, R

ML & Data: Scikit-learn, Azure Machine Learning, Snowflake, Streamlit

Cloud & Engineering: Azure, AWS, Docker, Kubernetes/AKS, Azure DevOps, Git, Linux

PROFESSIONAL EXPERIENCE

ELEVATE CREDIT | Data Scientist II (promoted Nov 2025) | Hybrid **June 2024 - Present**

- Engineered end-to-end Azure ML pipelines to reparameterize 10 production predictive models monthly, reducing model refresh time from two weeks to one day and enabling quantitative drift monitoring. Led adoption of Azure ML engineering practices, including versioned data assets, registered environments and models, and reproducible training and deployment pipelines.
- Developed and productionized a risk of chargeoff (ROC) collections model, beating existing third-party vendor solution performance and allowing the company to move our collections models in-house saving \$100K+ annually.
- Developed and productionized an XGBoost credit-risk model that improved AUC by 13% over existing solution; built a modular ~1,200-line Snowflake SQL layer to transform raw XML into validated training datasets.
- Designed and deployed Dockerized model inference APIs on Azure Kubernetes Service, incorporating unit tests and standardized deployment workflows for production reliability.
- Built and maintained an internal Python package with 50+ reusable modules for feature engineering, validation, and lifecycle management; adopted by a 15-member team across 12+ production projects.
- Implemented a Snowflake feature store containing 50+ proprietary engineered attributes, replacing ad hoc notebook logic with reusable, governed features.

XYLEM INC. | Software Engineer Intern | Raleigh, NC **May 2022 - Aug 2022**

- Developed three Angular application screens for an enterprise data-lake product and integrated customer data using SQL and AWS.
- Worked in an Agile development environment using Jira for tracking.
- Built Python scripts supporting enterprise security analysis for 1,000+ employees.

UNC CHAPEL HILL - COMPUTER SCIENCE | Undergraduate Learning Assistant | Chapel Hill, NC **Aug 2022 - May 2023**

- Designed and taught an MVC architecture module in COMP 301 for 50+ students and provided Java debugging support focused on object-oriented design and software engineering principles. Assisted with exam question development and grading.

SELECTED TECHNICAL PROJECT

ARDUINO MIDI LIGHT SHOW | Personal Project **July 2025**

- Engineered a real-time, event-driven system using Python, multithreading, serial I/O, and embedded C++ to synchronize LED hardware with audio playback.
- Implemented a custom low-latency serial protocol and extensible architecture for dynamically mapping MIDI attributes to lighting patterns.
- Technical write-up and tutorial posted on Medium with video demo

EDUCATION, LEADERSHIP & CERTIFICATION

M.S. Analytics, NC State University | May 2024

B.S. Computer Science, Minor in Religious Studies, UNC Chapel Hill | May 2023

Leadership & Honors: Elected student team lead for U.S. Department of Education practicum; later selected as Elevate technical liaison for the 2026 practicum cohort; graduated with distinction from UNC Chapel Hill, member of the Honors college; HackNC 2022 Best Junior Hackers winner.

Certification: AWS Certified Cloud Practitioner